Sai Venkatesh Varun M.M.

+1 (617) 606 8263 saivarunmm@gmail.com GitHub LinkedIn

EDUCATION

Northeastern University, Boston, MA — Sept. 2025 – Present
Khoury College of Computer Sciences
Pursuing **Master's in Computer Science**, CGPA: 3.9/4.0
Relevant Courses:, Machine Learning, Web Develeopment, Essentials of Data Science, Programming Design Paradigm, Database Management

Vellore Institute of Technology (VIT), Chennai, India — Jun 2021 – May 2025
Bachelor of Technology in Computer Science, CGPA: 8.34/10.0
Relevant Courses:, Artificial Intelligence, Digital Image Processing, Data Structures & Algorithms, Foundations of Blockchain Technology, Cryptography & Network Security

TECHNICAL SKILLS

Languages:	Python, R, SQL, Java, C, C++, Matlab
Web Technologies:	Selenium Web Driver, BeautifulSoup, HTML5, CSS3, PHP, XML
Databases:	MySQL, Relational Databases
Frameworks & Tools:	ETL Pipelines, Data Visualization, Git/GitHub, JUnit, Java Swing, SpringBoot, Aiven

WORK EXPERIENCE

BOSCH — Jul 2024 – Dec 2024
ML Engineer Intern

- Designed and created an AI RAG ChatBot for BOSCH employees based on a LLAMA model supporting text-to-text, text-to-URL, and text-to-video conversions.
- Contributed to design, development, and testing of chatbot functionalities and implemented feedback from BOSCH clients.
- Led a team of four in the NLP and web-scraping portion of the project, coordinating communication and delegating tasks.

TECHNICAL PROJECTS

Streaming Analytics Data Warehouse — Sept 2025 – Dec 2025

- Architected a year-partitioned star schema in a cloud-hosted MySQL environment, applying advanced relational database design to optimize multi-year streaming data.
- Engineered an automated ETL pipeline using R and SQLite to extract, clean, and validate over 98,000 streaming transactions, replacing manual workflows with efficient chunked processing.
- Developed automated data analysis scripts and reports using R Markdown to surface clear business insights and data visualizations for stakeholders.

Event Scheduler — Sept 2025 – Dec 2025
Java-based Desktop Event Schedular using MVC Architecture

- Architected a full-featured desktop calendar replicating Google/iCal functionalities, applying the **Model–View–Controller (MVC)** pattern to ensure strict separation of concerns and modularity.
- Developed an interactive GUI using **Java Swing** to support dynamic event management, error detection, and date navigation.
- Achieved **90%+ code coverage** by implementing **JUnit** test suites to validate business logic and complex edge cases.

Energy Efficient Transfer Learning for LLMs — Jul 2024 – Dec 2024
Adaptive Fine-Tuning with Dynamic Pruning

- Proposed and implemented adaptive fine-tuning and dynamic pruning strategies to minimize energy consumption and computational overhead in Large Language Models.
- Benchmarked performance using OPL on terabyte-scale datasets (Wikitext2, PTB, C4), maintaining high model accuracy.
- Validated model robustness by rigorously testing perplexity scores, achieving viable performance even at **50% sparsity**.

CERTIFICATIONS & ACHIEVEMENTS

- Completed certification in **Google Cloud Digital Leader** (Aug 2024) [Credential].
- Completed certification in **AWS Cloud Practitioner** (Aug 2024) [Credential].
- Participated in JK Tyre Indian Karting Race 2023 organized by ISIEINDIA.

RESEARCH PAPERS

- Enhanced Sensitivity-Aware Mixed Sparsity Pruning for Energy-Efficient Large Language Models, Sustainable Computing: Informatics and Systems - Submitted
- Comparitive Analysis of Glove CNN, LSTM and BERT Models for Hate Speech Detection, Scientific Reports-Submitted